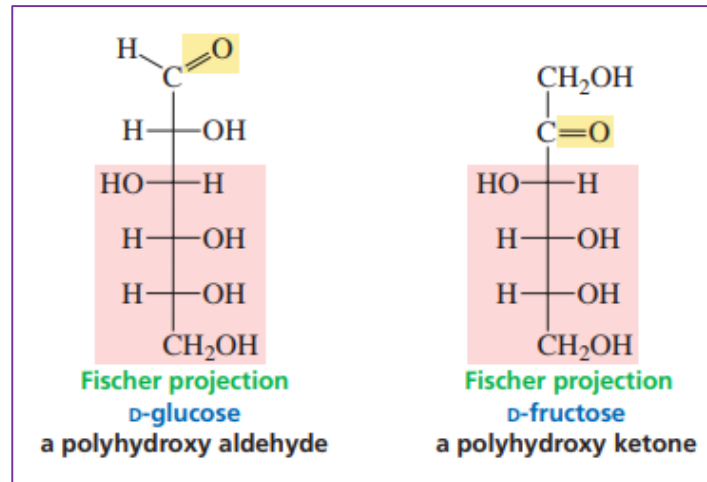


Carbohydrates

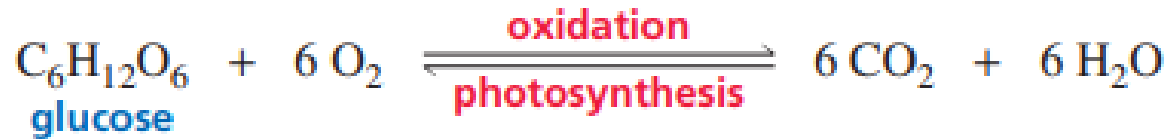
Carbohydrates

- **BIOORGANIC COMPOUNDS** ARE ORGANIC COMPOUNDS FOUND IN LIVING SYSTEMS.
- The first group of bioorganic compounds we will look at are **carbohydrates**—the **most abundant class** of compounds in the biological world, making up more than **50% of the dry weight** of the Earth's biomass.
- Early chemists noted that **carbohydrates have molecular formulas** that make them appear to be **hydrates of carbon**, $C_n(H_2O)_n$ —hence the name. Structural studies later revealed that **these compounds are not hydrates** because they do not contain **intact water** molecules. Nevertheless, the term **carbohydrate persisted**.
- **Carbohydrates** are **polyhydroxy aldehydes** such as **glucose**, polyhydroxy ketones such as **fructose**, and compounds such as **sucrose** formed by linking polyhydroxy aldehydes or polyhydroxy ketones together.
- The chemical structures of carbohydrates are commonly represented by Fischer projections. Notice that both glucose and fructose have the **molecular formula $C_6H_{12}O_6$** , consistent with the general formula $C_6(H_2O)_6$ that made early chemists think that these compounds were hydrates of carbon. Notice, too, that the **structures of glucose and fructose** differ only at the top two carbons.



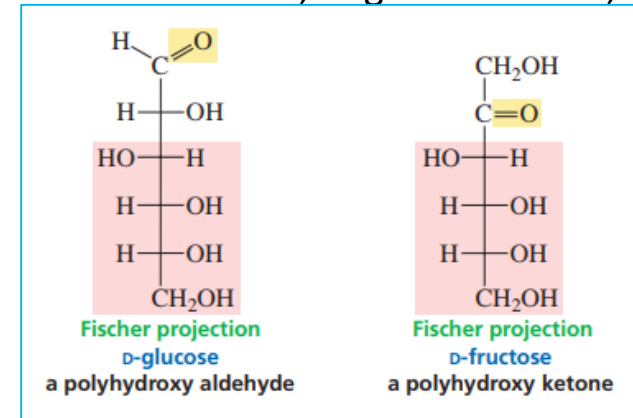
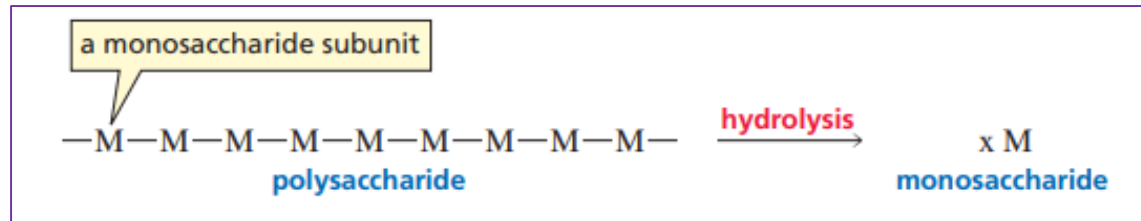
Carbohydrates

The most abundant carbohydrate in nature is glucose. Animals obtain glucose from food that contains glucose, such as plants. Plants produce **glucose by *photosynthesis***. During photosynthesis, plants take up water through their roots and use carbon dioxide from the air to synthesize glucose and oxygen. Because **photosynthesis is the reverse of the process used by organisms to obtain energy—specifically**, the oxidation of glucose to carbon dioxide and water—plants require energy to carry out photosynthesis. They obtain that energy **from sunlight**, which is captured by **chlorophyll molecules in green plants**. Photosynthesis uses the CO₂ that animals exhale as waste and generates the O₂ that animals inhale to sustain life. Nearly all the oxygen in the atmosphere has been released by photosynthetic processes.



CLASSIFICATION OF CARBOHYDRATES

- The terms **carbohydrate**, **saccharide**, and **sugar** are used interchangeably. **Saccharide** comes from the word for sugar in several early languages (*sarkara* in Sanskrit, *sakcharon* in Greek, and *saccharum* in Latin)
- Simple carbohydrates** are **monosaccharides** (single sugars); **complex carbohydrates** contain two or more monosaccharides linked together. **Disaccharides** have two monosaccharides linked together, **oligosaccharides** have 3 to 10 (*oligos* is Greek for “few”), and **polysaccharides** have 10 or more. Disaccharides, oligosaccharides, and polysaccharides can be broken down to monosaccharides by hydrolysis.

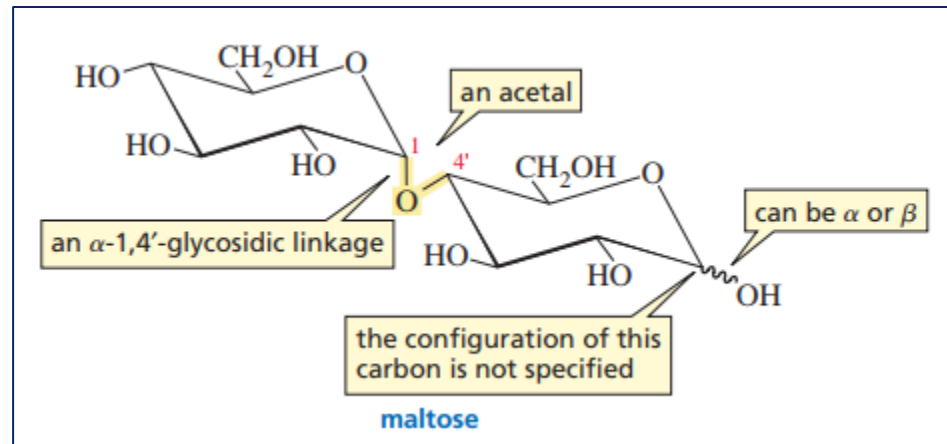


- A **monosaccharide** can be a polyhydroxy aldehyde such as glucose or a polyhydroxy ketone such as fructose. **Polyhydroxy aldehydes** are called **aldoses** (“ald” is for aldehyde; “ose” is the suffix for a sugar); **polyhydroxy ketones** are called **ketoses**.
- Monosaccharides** are also classified according to **the number of carbons they contain**: those with three carbons are **trioses**, those with four carbons are **tetroses**, those with five carbons are **pentoses**, and those with six and seven carbons are **hexoses** and **heptoses**. Therefore, a six-carbon polyhydroxy aldehyde such as **glucose** is an **aldohexose**, whereas a six-carbon polyhydroxy ketone such as **fructose** is a **ketohehexose**.

CLASSIFICATION OF CARBOHYDRATES

DISACCHARIDES

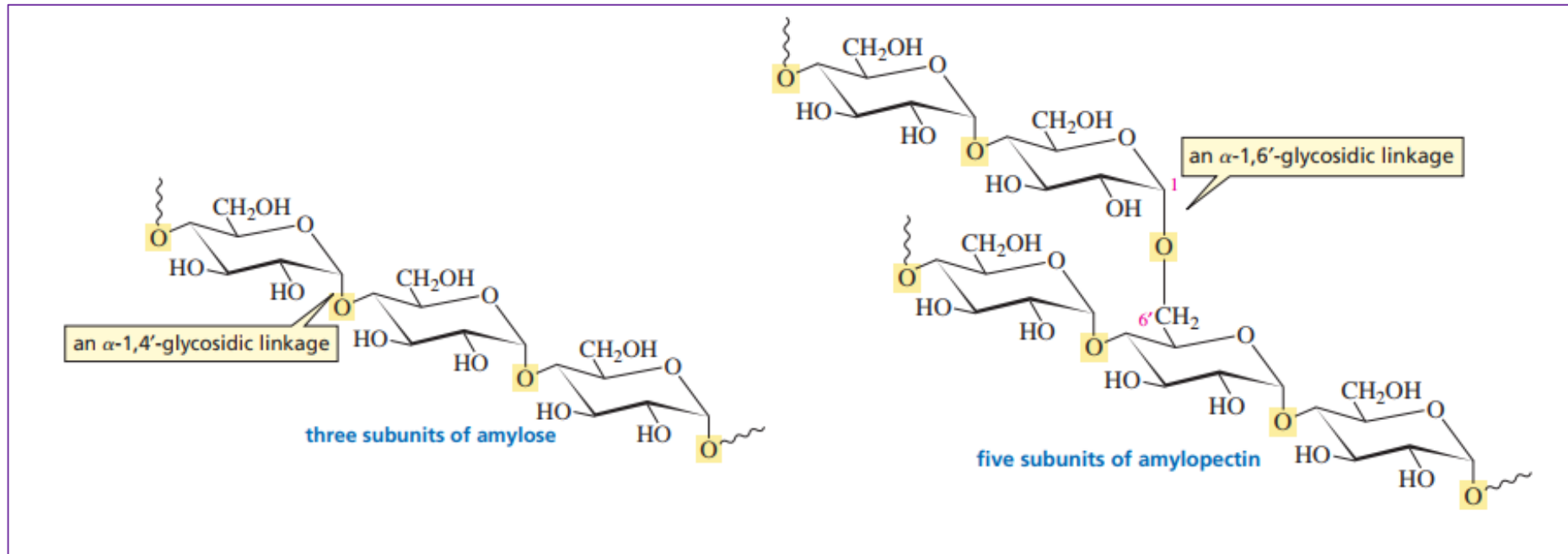
- **Disaccharides** are compounds that consist of **two monosaccharide subunits** hooked together by a **glycosidic linkage**.
- For example, **maltose**, a **disaccharide** obtained from the hydrolysis of starch, contains **two D-glucose** subunits connected by a **glycosidic** linkage. This particular linkage is called an **α -1,4-glycosidic linkage** because the linkage is between **C-1 of one sugar** subunit and **C-4 of the other**, and the oxygen bonded to the **anomeric** carbon in the **glycosidic** linkage is in the **α -position**. The prime superscript indicates that C-4 is not in the same ring as C-1.



CLASSIFICATION OF CARBOHYDRATES

POLYSACCHARIDES

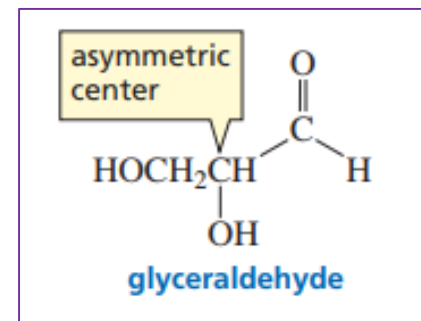
Polysaccharides contain as few as **10** or as many as **several thousand monosaccharide** units joined together by **glycosidic** linkages. The most common polysaccharides are **starch** and **cellulose**. Starch is the major component of flour, potatoes, rice, beans, corn, and peas. It is a mixture of two different polysaccharides: **amylose (~20%)** and **amylopectin (~80%)**. **Amylose** is composed of unbranched chains of **D -glucose units** joined by **α -1,4- glycosidic** linkages.



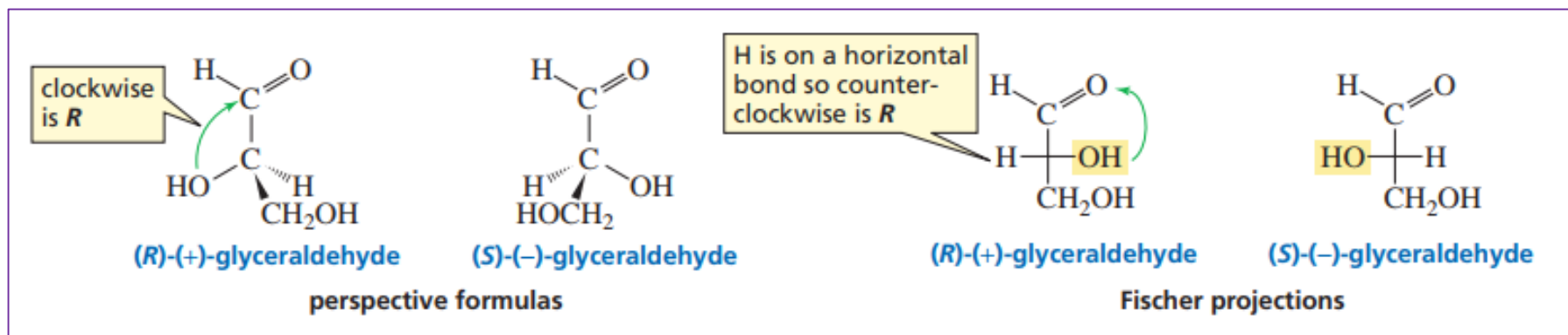
CLASSIFICATION OF CARBOHYDRATES

THE D AND L NOTATION

The smallest aldose, and the only one whose name does not end in “ose,” is **glyceraldehyde**, an aldotriose.



Because glyceraldehyde has an asymmetric center, **it can exist as a pair of enantiomers**. We know that the isomer on the left has the **R configuration** because an arrow drawn from the highest priority substituent (**OH**) to the next highest priority substituent (**HC=O**) is clockwise and the lowest priority group is on a hatched wedge. The *R* and *S* enantiomers drawn as Fischer projections are shown on the right.

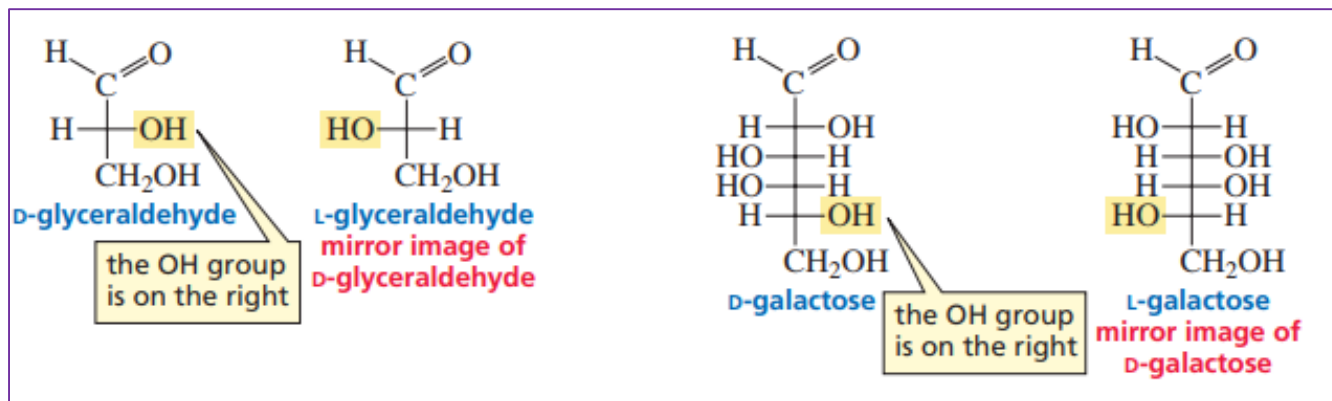


The notations **D** and **L** are used to describe the configurations of carbohydrates. In a Fischer projection of a monosaccharide, the carbonyl group is always **placed on top** (in the case of aldoses) or **as close to the top** as possible (in the case of ketoses).

CLASSIFICATION OF CARBOHYDRATES

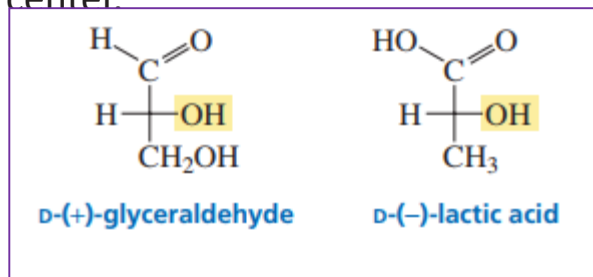
THE D AND L NOTATION

Examine the Fischer projection of **galactose** shown below and note that the compound has **four asymmetric centers** (C-2, C-3, C-4, and C-5). *If the OH group attached to **the bottommost asymmetric center** (the carbon second from the bottom) is on the right, **then the compound is a D-sugar**. If that same OH group is on the left, then the compound is an **L-sugar**.* Almost all sugars found in nature are D -sugars. The mirror image of a D -sugar is an L –sugar.



Like *R* and *S*, the symbols D and L indicate the **configuration of an asymmetric center**, but they **do not indicate** whether the compound rotates the plane of polarization of polarized light to the right (+) or to the left (-).

For example, D-glyceraldehyde is **dextrorotatory**, whereas D-lactic acid is **levorotatory**. In other words, optical rotation, like melting points or boiling points, is a physical property of a compound, whereas “*R*, *S*, D , and L ” are conventions humans use to indicate the configuration about an asymmetric center.

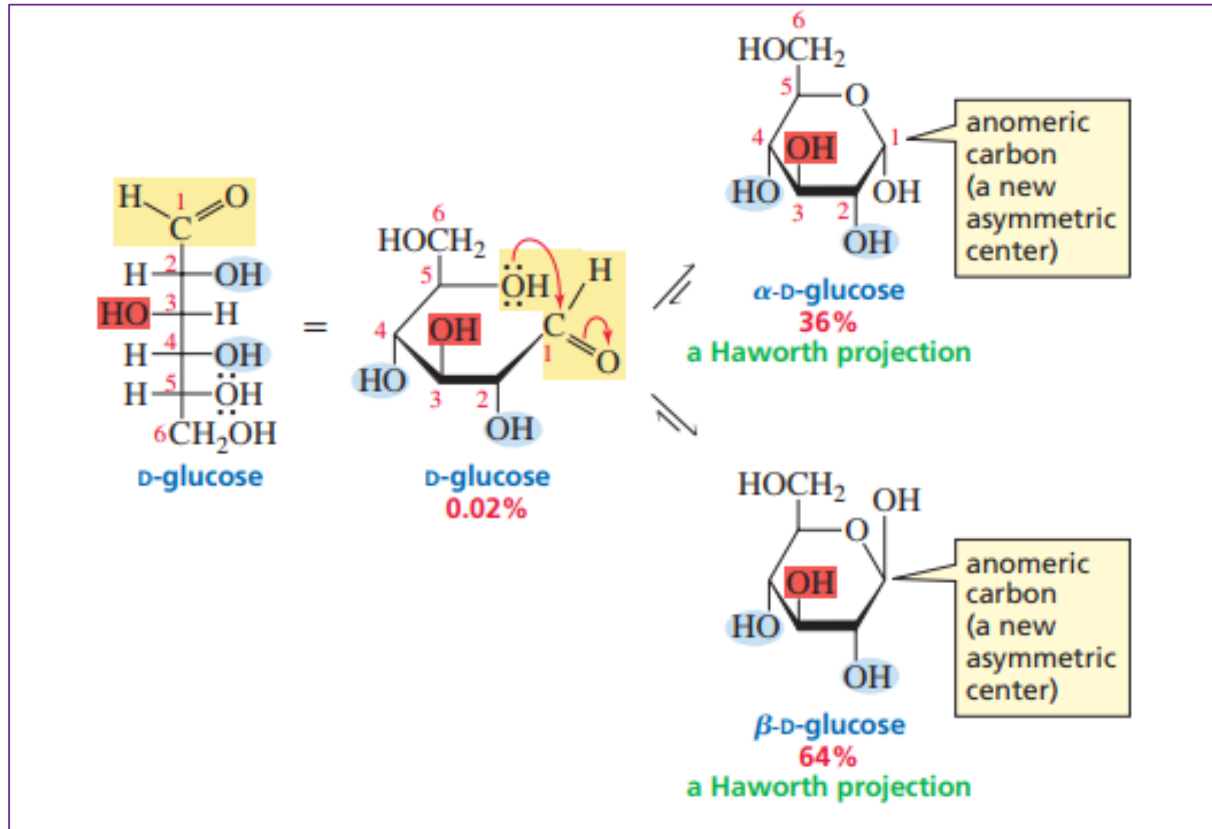


CARBOHYDRATES

MONOSACCHARIDES FORM CYCLIC HEMIACETALS

D-Glucose exists in three different forms: **the open-chain form** of D-glucose that we have been discussing and **two cyclic forms**- α - D-glucose and β - D-glucose.

We know that the two cyclic forms are different **because they have different melting points and different specific rotations.**



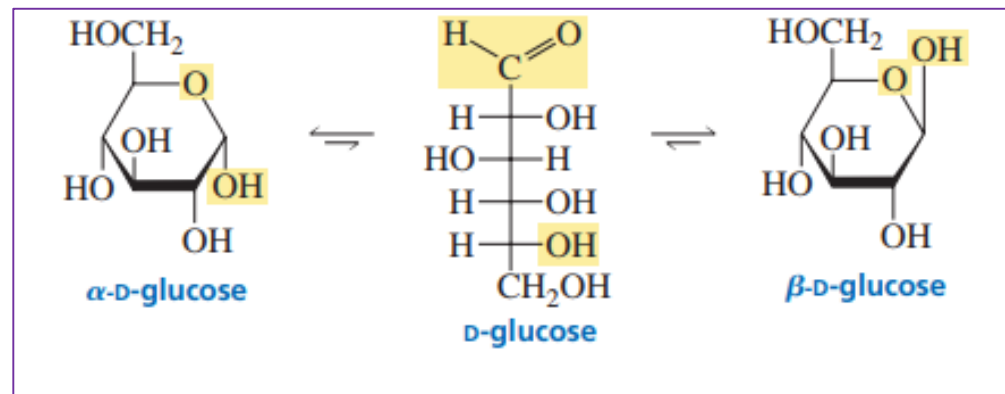
The cyclic hemiacetals shown here are drawn as Haworth projections. In a **Haworth projection**, the six-membered ring is represented as **flat and is viewed edge-on**. The ring oxygen is always placed in the back right-hand corner of the ring, with **C-1 on the right-hand side**, and the primary alcohol group attached to **C-5 is drawn up** from the back left-hand corner.

CARBOHYDRATES

MONOSACCHARIDES FORM CYCLIC HEMIACETALS

There are two different cyclic hemiacetals because the carbonyl carbon of the open chain aldehyde becomes a new asymmetric center in the cyclic hemiacetal. If the **OH group bonded to the new asymmetric center is down** (trans to the primary alcohol group at C-5), then the hemiacetal is **α - D -glucose**; if the **OH group is up** (cis to the primary alcohol group at C-5), then the hemiacetal is **β -D-glucose**. The mechanism for cyclic hemiacetal formation is the same as the mechanism for hemiacetal formation between individual aldehyde and alcohol molecules.

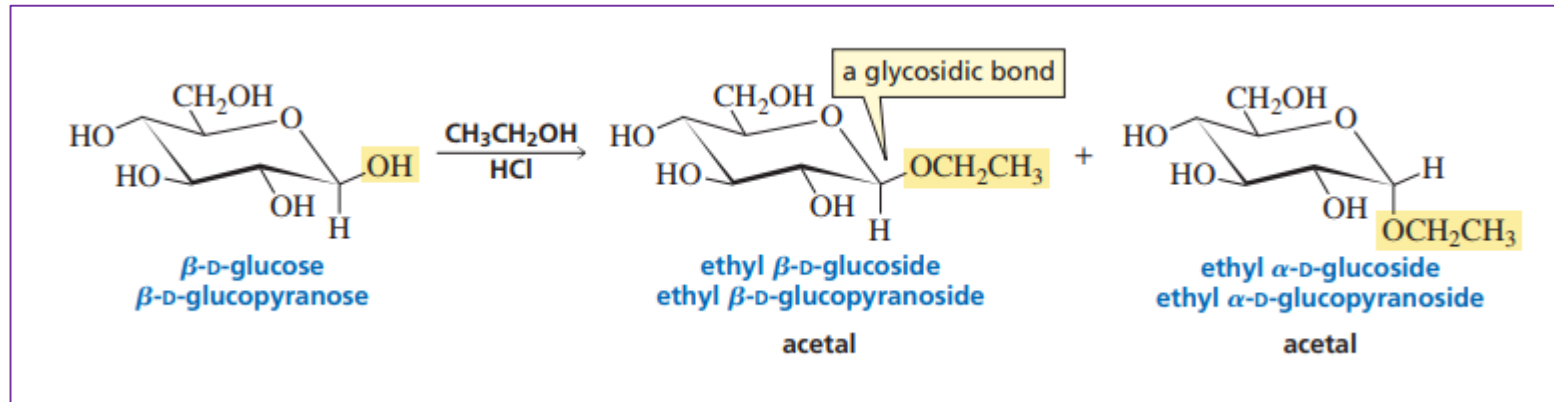
α - D -glucose and **β -D-glucose** are anomers. **Anomers** are two sugars that **differ in configuration only at the carbon** that was the **carbonyl carbon** in the open-chain form. This carbon is called the **anomeric carbon**. The prefixes **α -** and **β -** denote the configuration about the anomeric carbon. **Because anomers, like epimers, differ in configuration at only one carbon, they too are a particular kind of diastereomer.** Notice that the anomeric carbon is the only carbon in the molecule that is bonded to two oxygens.



CARBOHYDRATES

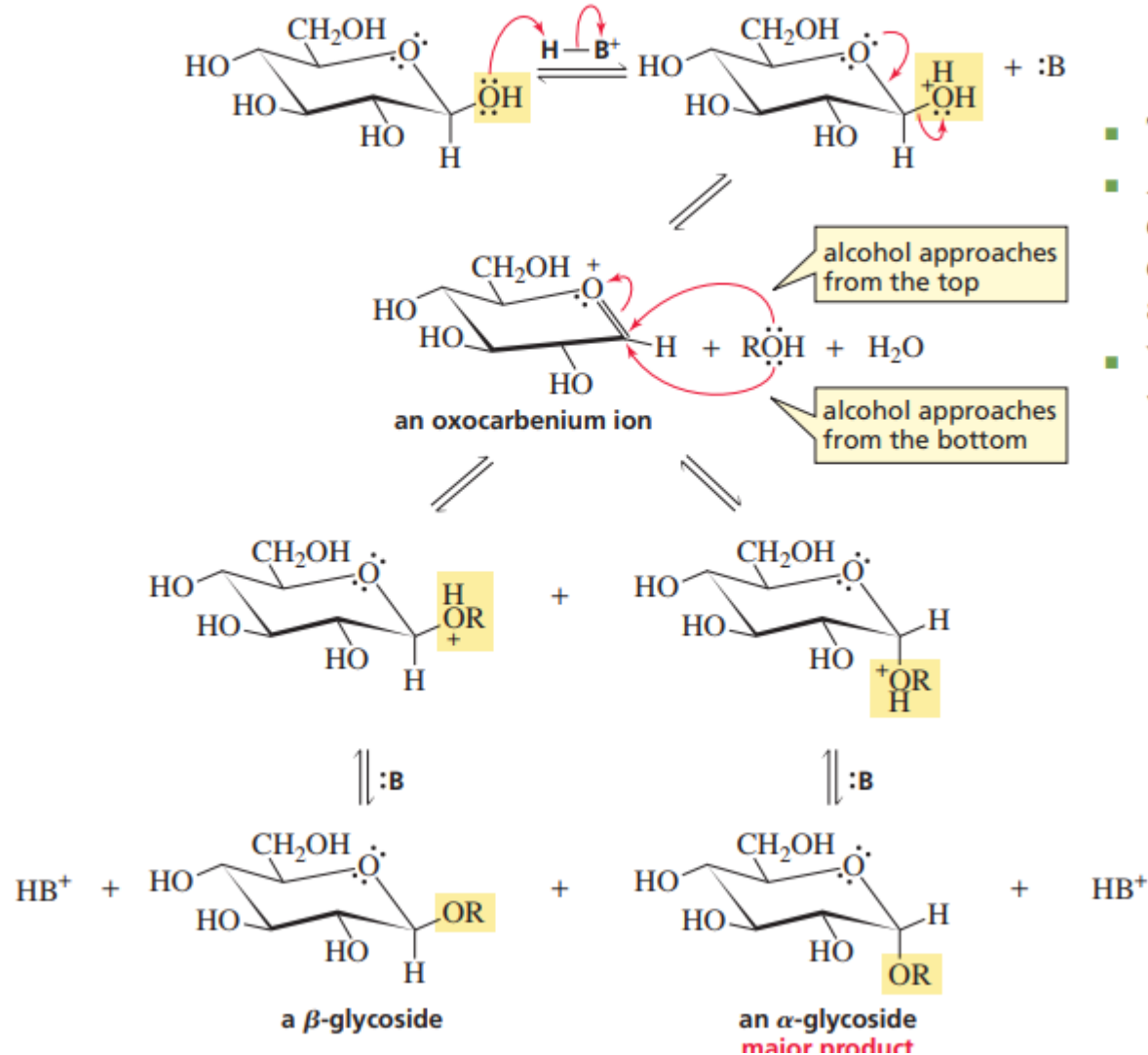
FORMATION OF GLYCOSIDES

In the same way that a hemiacetal reacts with an alcohol to form an acetal, the cyclic hemiacetal formed by a monosaccharide can react with an alcohol **to form two acetals**.



The **acetal** of a sugar is called a **glycoside**, and the bond between the **anomeric carbon and the alkoxy oxygen** is called a **glycosidic bond**. Glycosides are named by **replacing the “e” ending of the sugar’s name with “ide.”** Thus, a glycoside of glucose is a glucoside, a glycoside of **galactose** is a **galactoside**, and so on. If the pyranose or furanose name is used, the acetal is called a **pyranoside** or a **furanoside**.

MECHANISM FOR GLYCOSIDE FORMATION



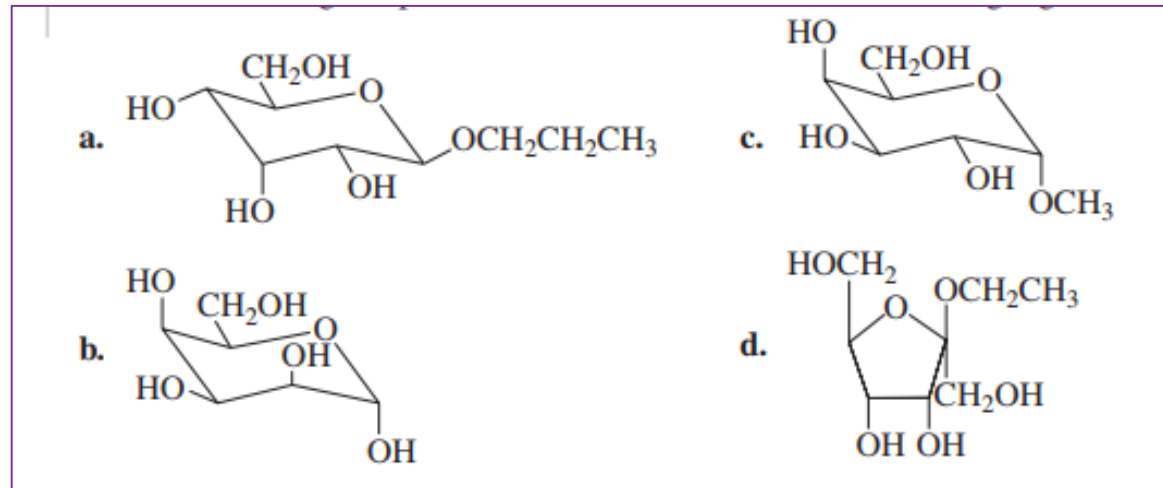
- The acid protonates the OH group bonded to the anomeric carbon.
- A lone pair on the ring oxygen helps eliminate a molecule of water. The anomeric carbon in the resulting oxocarbenium ion is sp^2 hybridized, so that part of the molecule is planar. (An **oxocarbenium** ion has a positive charge that is shared by a carbon and an oxygen.)
- When the alcohol approaches from the top of the plane, the β -glycoside is formed; when it approaches from the bottom of the plane, the α -glycoside is formed.

CARBOHYDRATES

REDUCING AND NONREDUCING SUGARS

Because glycosides are acetals, they **are not in equilibrium with the open-chain aldehyde (or ketone) in aqueous solutions**. Without being in equilibrium with a compound that has a carbonyl group, **they cannot be oxidized by Ag^+** . **Glycosides**, therefore, are **non-reducing sugars**—they cannot reduce Ag^+ .

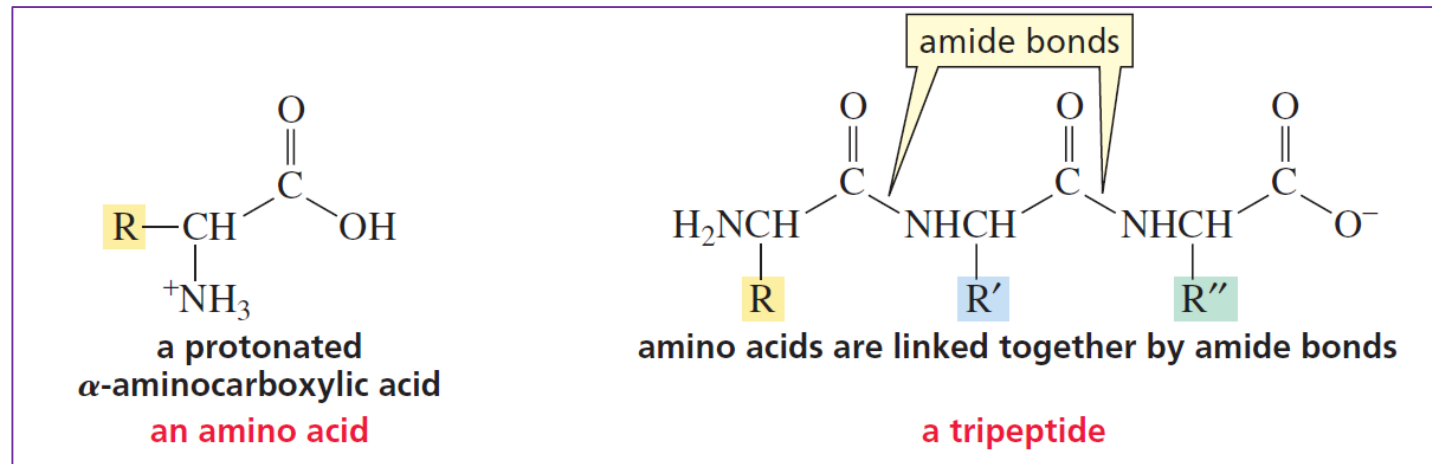
Hemiacetals, on the other hand, are in equilibrium with the open-chain sugars in aqueous solution, so they can reduce Ag^+ . In summary, as long as a sugar has an aldehyde, a ketone, or a hemiacetal group, it can reduce Ag^+ and therefore is classified as a **reducing sugar**. An acetal is a **non-reducing sugar**.



Amino Acids, Peptides, and Proteins

Amino Acids, Peptides, and Proteins

- An **amino acid** is a carboxylic acid with a protonated amino group on the α -carbon.
- **Peptides** and **proteins** are polymers of amino acids.
- The amino acids are linked together by amide bonds.



Amino acid polymers can be composed of any number of amino acids. A **dipeptide** contains two amino acids linked together, a **tripeptide** contains three, an **oligopeptide** contains 4 to 10, and a **polypeptide** contains many. **Proteins** are naturally **occurring polypeptides made up of 40 to 4000** amino acids. Proteins serve many functions in biological systems

Amino Acids, Peptides, and Proteins

Examples of the Diverse Functions of Proteins in Biological Systems	
Structural proteins	These proteins impart strength to biological structures or protect organisms from their environment. For example, collagen is the major component of bones, muscles, and tendons; keratin is the major component of hair, hooves, feathers, fur, and the outer layer of skin.
Protective proteins	Snake venoms and plant toxins are proteins that protect their owners from predators. Blood-clotting proteins protect the vascular system when it is injured. Antibodies and peptide antibiotics protect us from disease.
Enzymes	Enzymes are proteins that catalyze the reactions that occur in cells.
Hormones	Some hormones, compounds that regulate the reactions that occur in living systems, are proteins.
Proteins with physiological functions	These proteins include those that transport and store oxygen in the body, store oxygen in the muscles, and contract muscles.

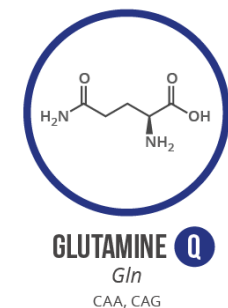
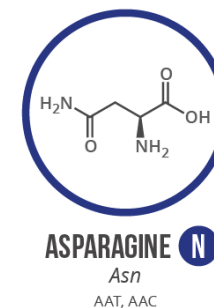
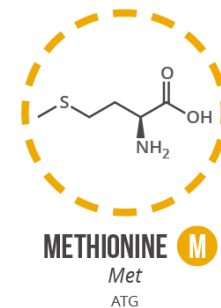
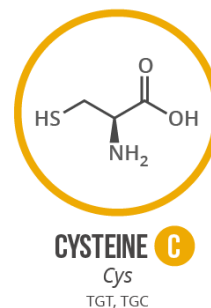
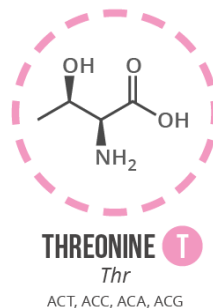
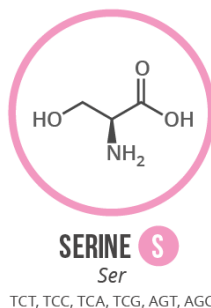
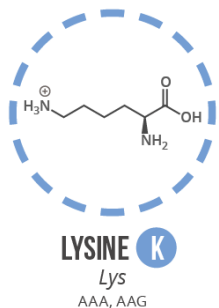
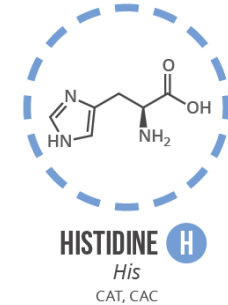
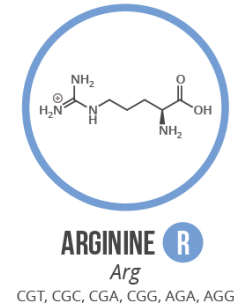
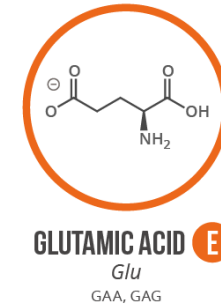
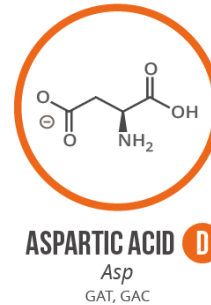
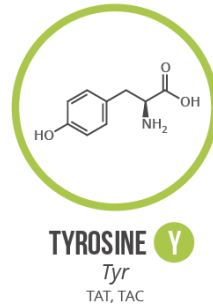
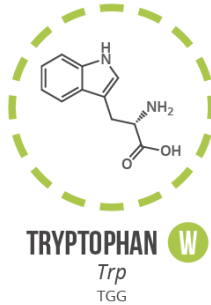
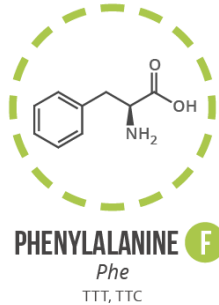
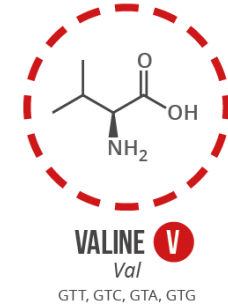
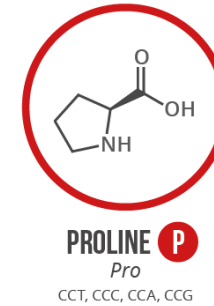
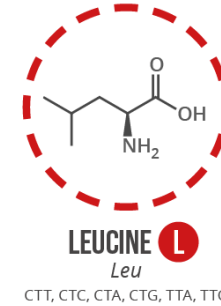
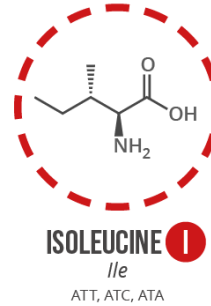
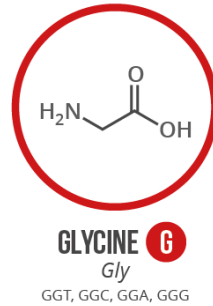
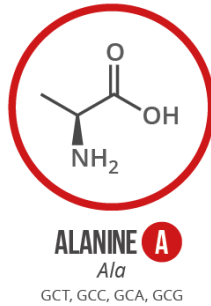
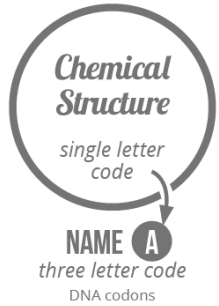
Proteins can be classified as either **fibrous or globular**. **Fibrous proteins** contain long chains of polypeptides arranged in **threadlike bundles**; these **proteins are insoluble in water**. **Globular proteins** tend to have roughly **spherical shapes** and most are **soluble in water**. All structural proteins are fibrous proteins; most enzymes are globular proteins.

Amino Acids, Peptides, and Proteins

Twenty naturally occurring Amino Acids

AMINO ACIDS ARE THE BUILDING BLOCKS OF PROTEINS IN LIVING ORGANISMS. THERE ARE OVER 500 AMINO ACIDS FOUND IN NATURE - HOWEVER, THE HUMAN GENETIC CODE ONLY DIRECTLY ENCODES 20. 'ESSENTIAL' AMINO ACIDS MUST BE OBTAINED FROM THE DIET, WHILST NON-ESSENTIAL AMINO ACIDS CAN BE SYNTHESISED IN THE BODY.

Chart Key: ● ALIPHATIC ● AROMATIC ● ACIDIC ● BASIC ● HYDROXYLIC ● SULFUR-CONTAINING ● AMIDIC ○ NON-ESSENTIAL ○ ESSENTIAL



Amino Acids, Peptides, and Proteins

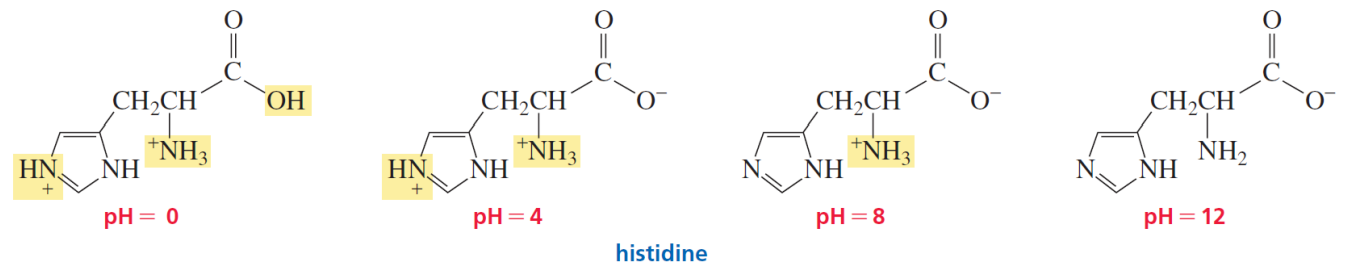
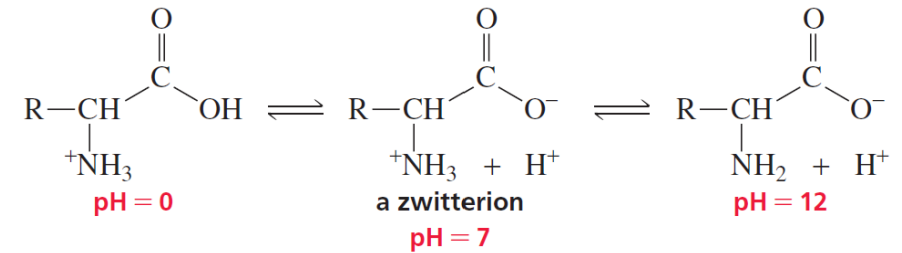
THE ACID-BASE PROPERTIES OF AMINO ACIDS

Every amino acid has a carboxyl group and an amino group, and each group can exist in an acidic form or a basic form, depending on the pH of the solution in which the amino acid is dissolved.

We have seen that compounds exist primarily in their acidic forms (that is, with their protons attached) in solutions that **are more acidic than** their pK_a values, and primarily in their basic forms (that is, without their protons) in solutions that **are more basic than** their pK_a values.

Notice that an amino acid can never exist **as an uncharged compound**, regardless of the pH of the solution. To be uncharged, an amino acid would have to lose a proton from an $^+NH_3$ group with a pK_a of about 9 before it would lose a proton from a COOH group with a pK_a of about 2. This is impossible because a weak acid ($pK_a = 9$) cannot lose a proton easier than a strong acid ($pK_a = 2$) can. Therefore, at physiological pH (7.4), an amino acid such as alanine exists as a dipolar ion, called a **zwitterion**. A **zwitterion** is a compound **that has a negative charge on one atom and a positive charge on a nonadjacent atom**. (The name comes from *zwitter*, German for “hermaphrodite” or “hybrid.”)

A few amino acids have side chains with ionizable hydrogens. The protonated imidazole side chain of **histidine**, for example, has a pK_a of 6.04. **Histidine**, therefore, can exist **in four different forms**, and the form that predominates depends on the pH of the solution.



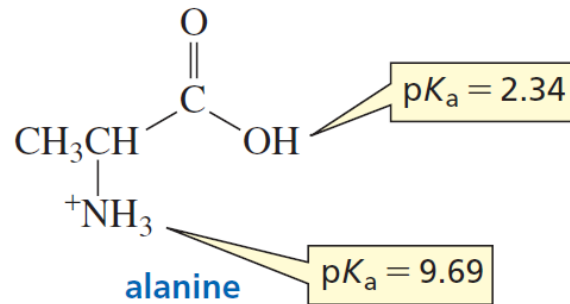
Amino Acids, Peptides, and Proteins

THE ISOELECTRIC POINT

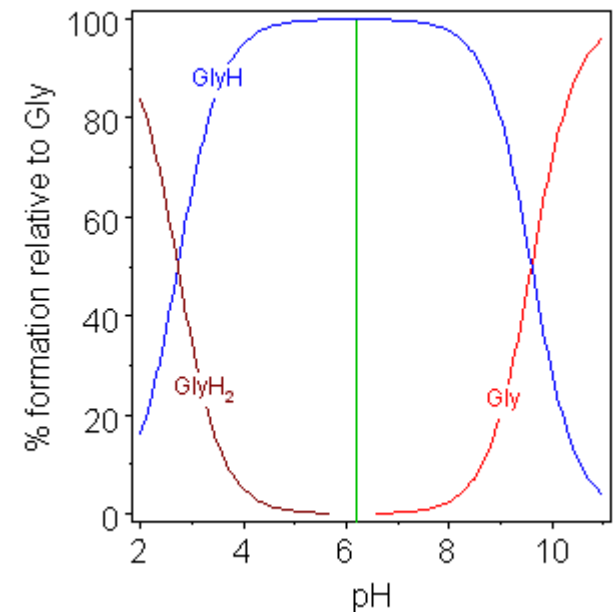
The **isoelectric point** (pI) of an amino acid is the pH at which it has **no net charge**. In other words, it is the pH at which the amount of positive charge on an amino acid exactly balances the amount of negative charge:

pI = pH at which there is no net charge

The pI of an amino acid that does *not have an ionizable side chain*—such as alanine—is midway between its two pK_a values. This is because half of the molecules have a negatively charged carboxyl group and half have an uncharged carboxyl group at $pH = 2.34$, whereas half of the molecules have a positively charged amino group and half have an uncharged amino group at $pH = 9.69$. As the pH increases from 2.34, the carboxyl group of more molecules becomes negatively charged; as the pH decreases from 9.69, the amino group of more molecules becomes positively charged. Therefore, the number of negatively charged groups equals the number of positively charged groups at the intersection (average) of the two pK_a values.



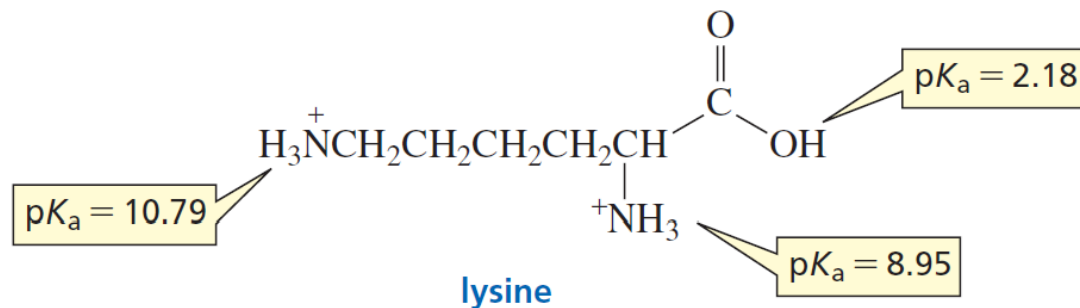
$$pI = \frac{2.34 + 9.69}{2} = \frac{12.03}{2} = 6.02$$



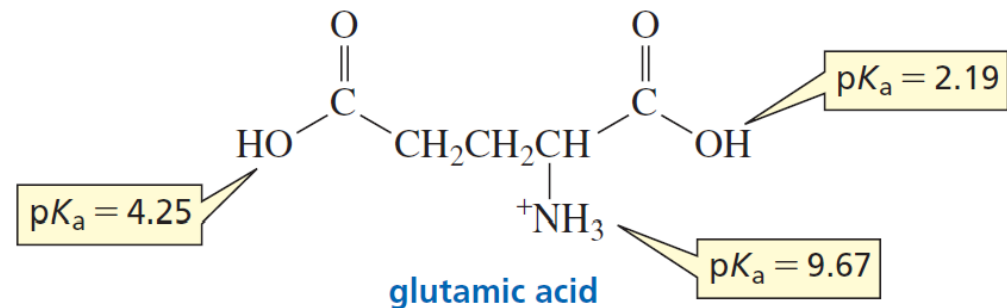
Amino Acids, Peptides, and Proteins

THE ISOELECTRIC POINT

The **pI** of most amino acids that *have an ionizable side chain* is the average of the **pK_a** values of **the similarly ionizing groups** (either positively charged groups ionizing to uncharged groups or uncharged groups ionizing to negatively charged groups). For example, the **pI** of lysine is the average of the **pK_a** values of the two groups that are positively **charged in their acidic form and uncharged in their basic form**. The pI of glutamic acid, on the other hand, is the average of the **pK_a** values of the two groups that are **uncharged in their acidic form and negatively charged in their basic form**.



$$\text{pI} = \frac{8.95 + 10.79}{2} = \frac{19.74}{2} = 9.87$$

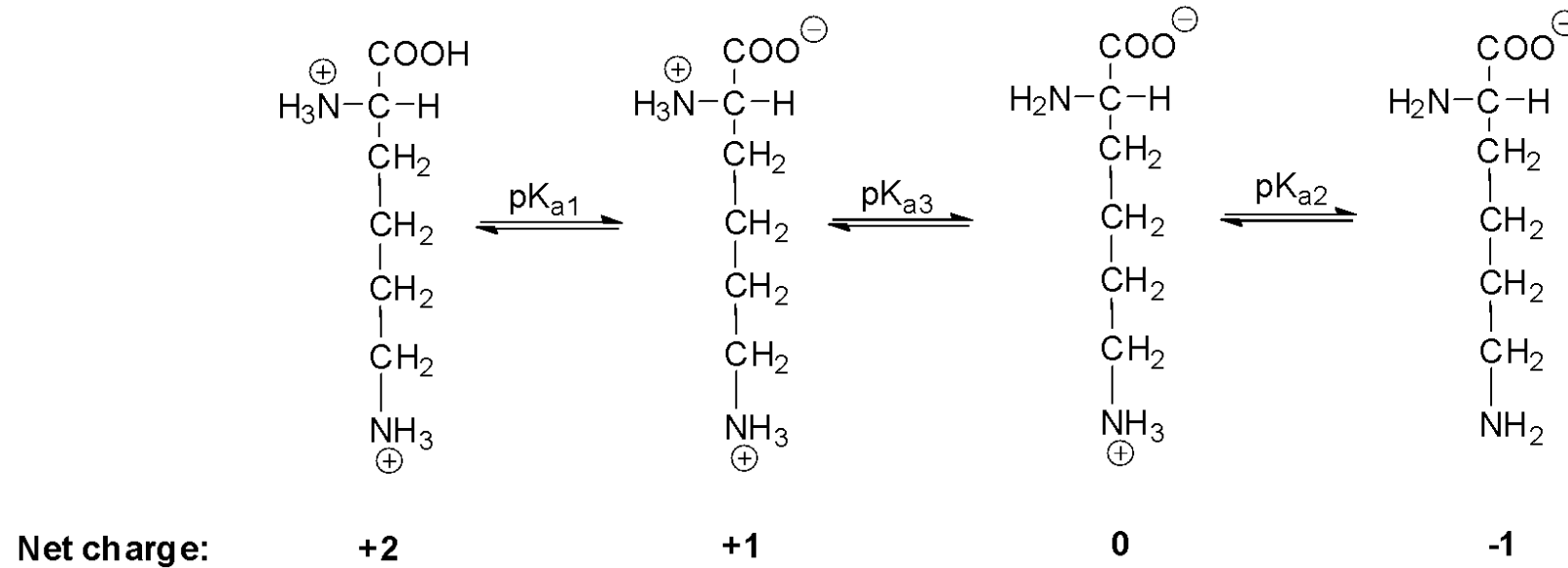


$$\text{pI} = \frac{2.19 + 4.25}{2} = \frac{6.44}{2} = 3.22$$

Amino Acids, Peptides, and Proteins

THE ISOELECTRIC POINT

Lysine



Since the isoelectric point is given by the average of the pK_a values **that involve the zwitterion**, so we can write the formula for lysine as:

$$pI = (pK_{a3} + pK_{a2}) \text{ divided by } 2$$

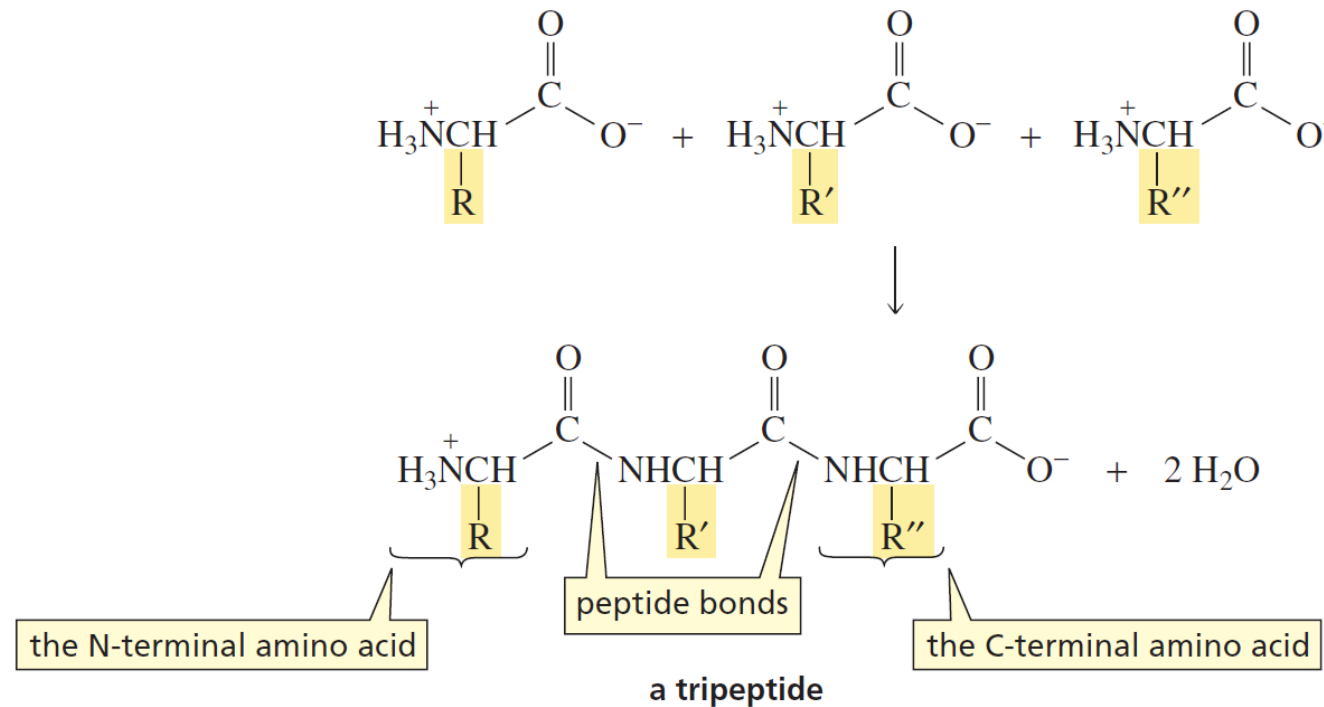
Amino Acids, Peptides, and Proteins

PEPTIDE BONDS AND DISULFIDE BONDS

Peptide bonds and disulfide bonds are the only covalent bonds that join amino acids together in a peptide or a protein.

Peptide Bonds

The amide bonds that link amino acids are called **peptide bonds**. By convention, peptides and proteins are written with the free amino group (of the **N-terminal amino acid**) on the left and the free carboxyl group (of the **C-terminal amino acid**) on the right.



Amino Acids, Peptides, and Proteins

What is the peptide bond?

Amino acids are joined together to form proteins. The covalent bond between two amino acids is the result of a condensation reaction (where water is released, as shown in Figure 3) and is called the *peptide bond*. Two amino acids joined together form a *dipeptide* and a longer chain of amino acids is called a *polypeptide*, or sometimes simply *peptide*.

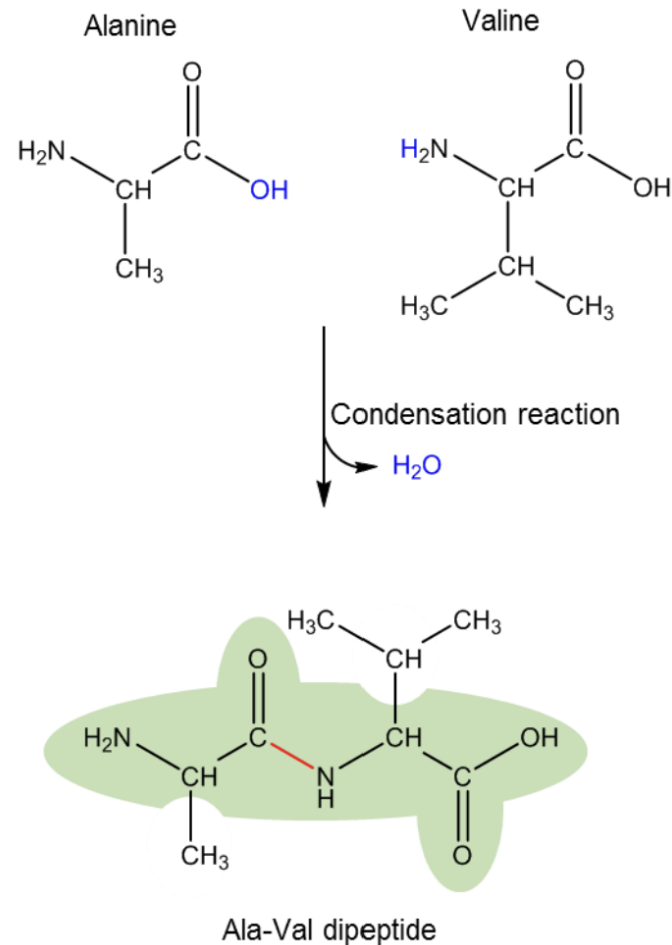
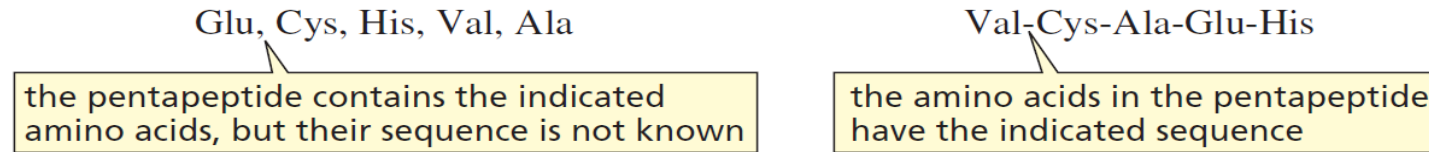


Figure 3 | Formation of an alanine-valine dipeptide. A water molecule (blue) is released by the condensation reaction and the peptide bond (red) is formed. The backbone is highlighted in green.

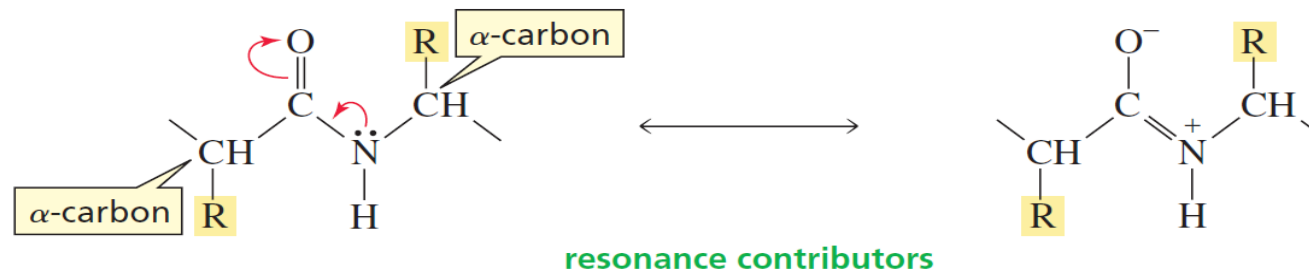
Amino Acids, Peptides, and Proteins

When the identities of the amino acids in a peptide are known but their sequence is not, the amino acids are written separated by commas. When their sequence is known, the amino acids are written connected by hyphens. For example, in the pentapeptide represented on the right, valine is the N-terminal amino acid and histidine is the C-terminal amino acid. The amino acids are numbered starting with the N-terminal end. Alanine is therefore referred to as Ala 3 because it is the third amino acid from the N-terminal end.



In naming a peptide, adjective names (ending in “yl”) are used for all the amino acids except the C-terminal amino acid. Thus, the preceding pentapeptide is valylcysteinylalanylglutamylhistidine. Each amino acid has the L configuration unless otherwise specified.

A peptide bond has about 40% double-bond character because of electron delocalization. Steric strain in the cis configuration causes the trans configuration about the amide linkage to be more stable, so the α -carbons of adjacent amino acids are trans to each other.

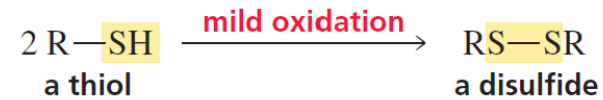


Amino Acids, Peptides, and Proteins

PEPTIDE BONDS AND DISULFIDE BONDS

Disulfide Bonds

When thiols are oxidized under mild conditions, they form a **disulfide**—a compound with an S—S bond. (Like C—H bonds, the number of S—H bonds decreases in an oxidation reaction and increases in a reduction reaction.)

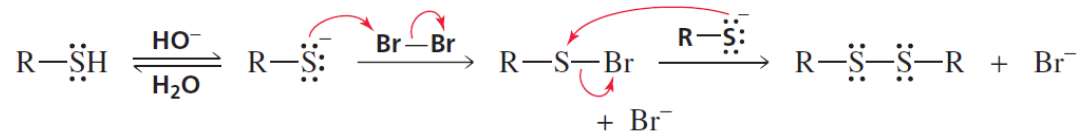


An oxidizing agent commonly used for this reaction is Br₂ (or I₂) in a basic solution.

Amino Acids, Peptides, and Proteins

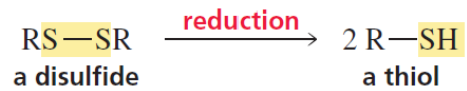
PEPTIDE BONDS AND DISULFIDE BONDS

MECHANISM FOR THE OXIDATION OF A THIOL TO A DISULFIDE

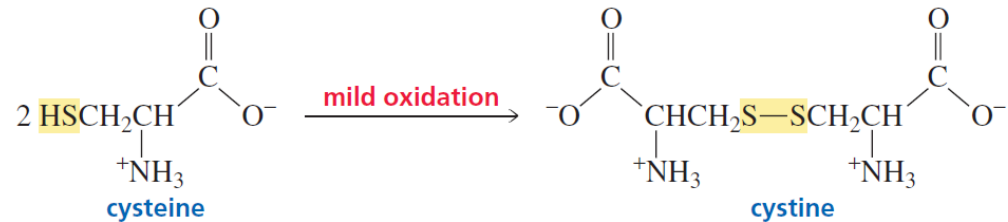


- A thiolate ion attacks the electrophilic bromine of Br₂.
- A second thiolate ion attacks the sulfur and eliminates Br⁻.

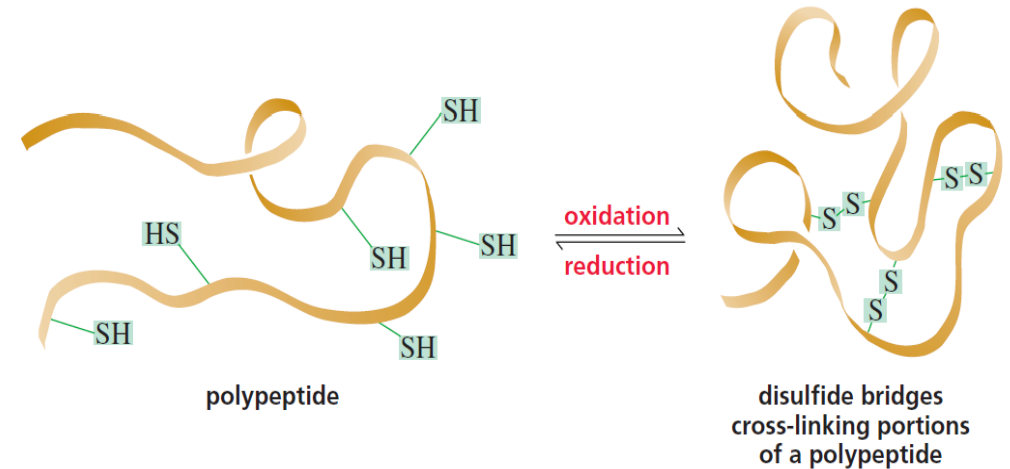
Because thiols can be oxidized to disulfides, disulfides can be reduced to thiols.



The amino acid cysteine contains a thiol group, so two cysteine molecules can be oxidized to a disulfide. The disulfide is called cystine.



Two cysteines in a protein can be oxidized to a disulfide, creating a bond known as a **disulfide bridge**. Disulfide bridges are the only covalent bonds that are found between nonadjacent amino acids in peptides and proteins. They contribute to the overall shape of a protein by linking cysteines found in different parts of the peptide backbone, as shown in Figure 19.10.



Amino Acids, Peptides, and Proteins

AN INTRODUCTION TO PROTEIN STRUCTURE

Proteins are described by four levels of structure, called primary, secondary, tertiary, and quaternary.

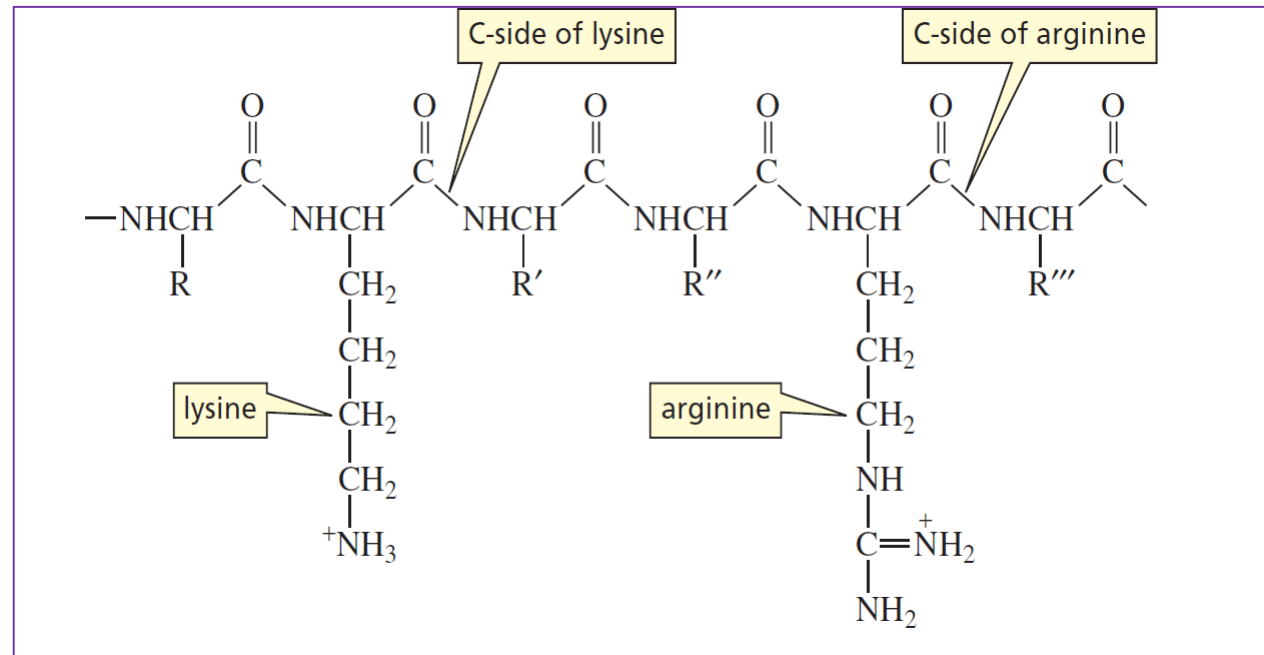
- The **primary structure** of a protein is the sequence of the amino acids in the chain and the location of all the disulfide bridges.
- **Secondary structures** are regular conformations assumed by segments of the protein's backbone when it folds.
- The **tertiary structure** is the three-dimensional structure of the entire protein.
- If a protein has more than one polypeptide chain, it also has a quaternary structure. The **quaternary structure** is the way the individual polypeptide chains are arranged with respect to one another.

Amino Acids, Peptides, and Proteins

PRIMARY STRUCTURE

The primary structure of a protein is the **sequence of the amino acids that constitute it**.

Because of the nature of the peptide bond, the backbone of a polypeptide will have a **single primary amine** at one end and a **single carboxylic acid** at the other end (as they do not take part in a peptide bond). Those ends are called the **N-terminus** (primary amine) and the **C-terminus** (carboxylic acid). The sequence of a polypeptide is always read from the **N-terminus to the C-terminus**.



Amino Acids, Peptides, and Proteins

SECONDARY STRUCTURE

Secondary structure

The peptide bond between two residues is a single bond, but it is said to have a semi double-bond character. This means that it is particularly rigid for a single bond, forming a planar structure called the amide plane, as shown in Figure 5.

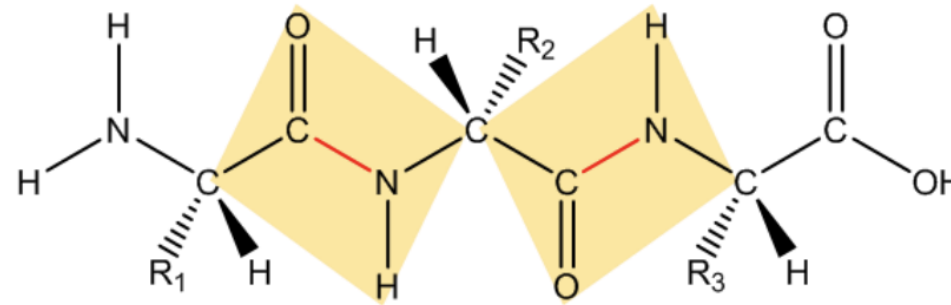


Figure 5 | The amide planes of a tripeptide. Each peptide bond (red) forms a planar structure, the amide plane (yellow), due to its semi double-bond character. R_{1-3} : side chains.

The angles between subsequent amide planes in a polypeptide are called *torsion angles*. They can only adopt certain values, and those values impose certain conformations (or folds) on the backbone.

⇒ The secondary structure of a protein is the local fold of the protein backbone.

Some of those local folds form precise, regular structures, often stabilised by hydrogen bonds. The two most common examples of secondary structure elements are α -helices and β -sheets.

Amino Acids, Peptides, and Proteins

SECONDARY STRUCTURE

In an α -helix, the polypeptide chain forms a right-handed helical structure with 3.6 residues per turn (Figure 6). The helix is stabilised by hydrogen bonds between the backbone N–H of each residue and the backbone C=O of the amino acid four residues earlier in the sequence. The core of the helix is tightly packed and all the side chains project outward.

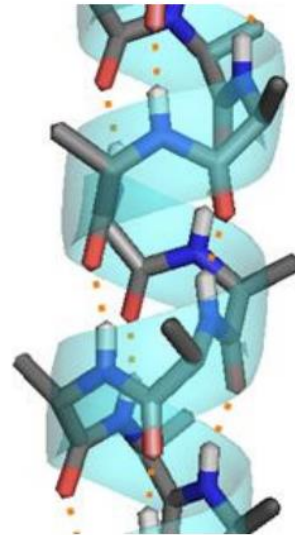


Figure 6 | Example of an α -helix containing alanine residues only. Hydrogen bonds (orange) between backbone atoms four residues apart stabilise the α -helix. Dark grey: carbon; blue: nitrogen; red: oxygen; white: hydrogen. Only the hydrogen atoms involved in hydrogen bonds are displayed.

Amino Acids, Peptides, and Proteins

SECONDARY STRUCTURE

β -sheets are the other most common type of secondary structure element, shown in Figure 7. They are also stabilised by hydrogen bonds, but between different chains, whereas in an α -helix, the hydrogen bonds are all within the same helix. Similarly to α -helices however, the hydrogen bonds stabilising β -sheets are between backbone N–H and C=O groups.

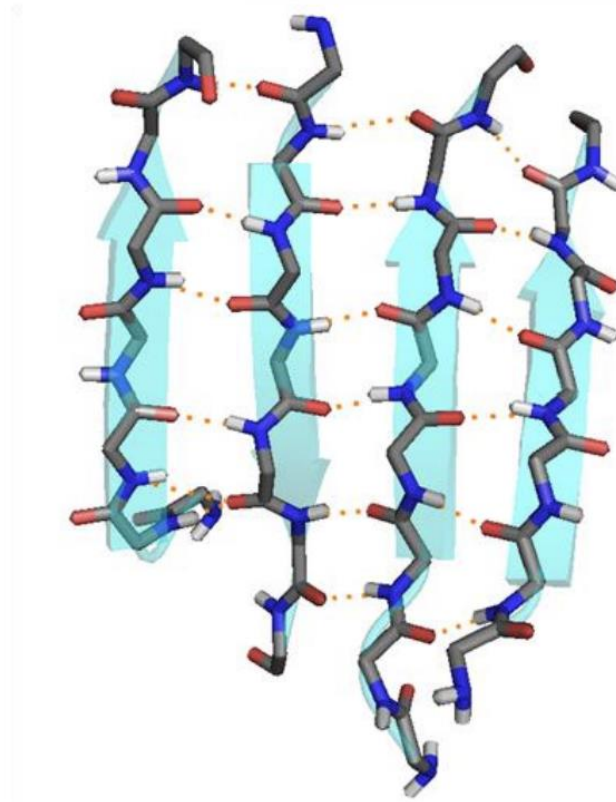


Figure 7 | Example of a β -sheet. All side chain atoms are hidden for simplicity. Hydrogen bonds (orange) between backbone atoms from adjacent β -strands stabilise the β -sheet. Dark grey: carbon; blue: nitrogen; red: oxygen; white: hydrogen. Only the hydrogen atoms involved in hydrogen bonds are displayed.

Amino Acids, Peptides, and Proteins

TERTIARY STRUCTURE

Tertiary structure

⇒ The tertiary structure of a protein is its overall 3D arrangement: the folding of secondary structure elements and the position of side chains.

The hydrophobic effect is responsible for most of the tertiary structure of a protein: it is energetically favourable for the protein to fold and bury its hydrophobic residues within its core, away from the surrounding water.

Other bonds and interactions also help the protein fold into the correct tertiary structure. Disulphide bonds are covalent bonds between the sulphur atoms of two cysteine residues. Salt bridges are electrostatic interactions between a negatively charged side chain and a positively charged one. Hydrogen bonds and van der Waals interactions (between hydrophobic residues) are also involved in the tertiary structure. All those forces, interactions and bonds are shown in Figure 8.

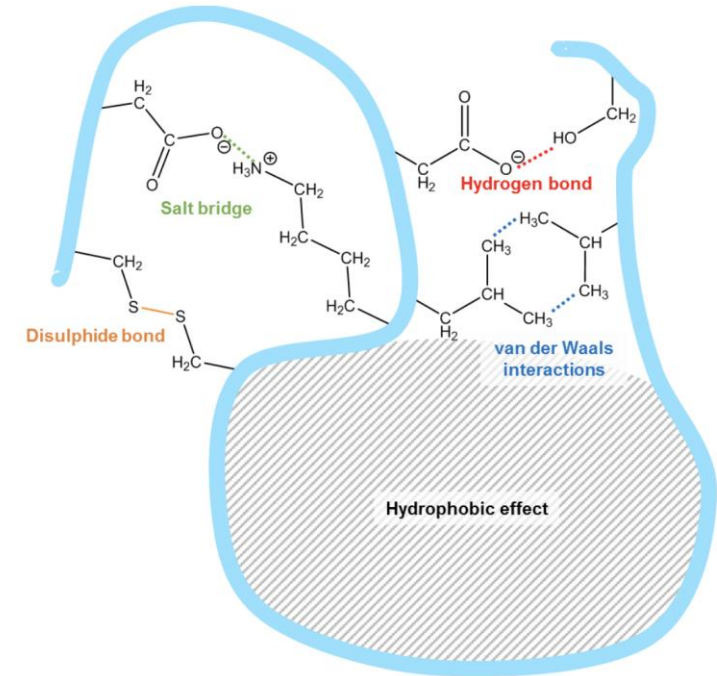


Figure 8 | The forces, bonds and interactions responsible for protein tertiary structure.

Amino Acids, Peptides, and Proteins

QUATERNARY STRUCTURE

Quaternary structure

Some proteins are made up of several polypeptide chains, which assemble once they have adopted their individual tertiary structures. The polypeptide chains may be identical or not: haemoglobin, for instance, has two copies of the same chain and two copies of another, different chain. Antibodies also contain four chains: two heavy chains and two light chains, as shown in Figure 9.

Some proteins are also covalently attached to a non-protein element, e.g. the haem cofactor in haemoglobin (cf. worksheet on haemoglobin).

⇒ The quaternary structure of a protein is the assembly of several polypeptide chains, and sometimes the addition of a non-protein element, to form a functional protein.

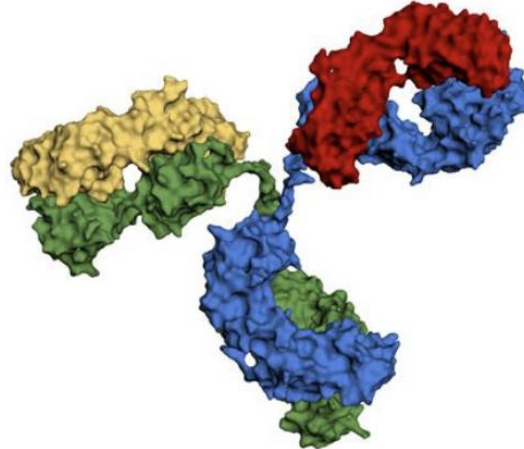


Figure 9 | The quaternary structure of an antibody, comprising two heavy chains (blue and green) and two light chains (yellow and red).

The same forces, bonds and interactions responsible for tertiary structure may be involved in holding different polypeptide chains together.